

ALIM YAKUB

RESEARCH ASSOCIATE

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EDUCATION

Newcastle University **2018 - 2019**
MSc Energy & Sustainability Distinction (Top Student Award)

Newcastle University **2016 - 2018**
BSc Mechanical Design & Manufacturing Engineering 2:1

EXPERTISE

- Python optimization
- Python machine learning
- 3D CAD modelling
- Data analytics and visualization
- Engineering & Human-centered design

EXPERIENCE

Research Associate **Jan 2023 - Sep 2023**

CNRS@CREATE

- Conversion of a MATLAB optimization program into Python
- Replacing an optimization controller of a grid connected distributed energy system into an adaptive data-driven (machine learning) lightweight compact linear tree controller

Research Associate **Apr 2022 - Sep 2022**

Surbana Jurong-NTU Corporate Lab

- Created a program for the generation and prediction of efficiency values for cooling systems using python machine learning regression algorithm
- Created a program for the optimization of chiller systems sizing and daily operations using python

Research Associate **Nov 2020 - Apr 2022**

NTU Institute of Science & Technology for Humanity

- Using python/javascript for data collection, analysis & generating data visualization of a network of NISTH fellow's publications in relation to the UN Sustainable development Goals on an online platform
- Assist in report writing & idea generation for new research topics related to science & technology with humanities angles
- Supervise FYP undergraduate on their social robot for elderly to promote inter-generational connectivity
- Building & design of NISTH 2021 Annual web report

Research Assistant **2018 - 2019**

Newcastle University

- Design, prototyping and fabrication of equipment needed for testing & experiments
- Calibration of prototype equipment
- Testing and processing of samples with prototype equipment

LANGUAGES

- English
- Mandarin
- Malay

RESEARCH INTEREST

- Scientific & social aspects of Sustainability
- Renewable energy generation
- Design
- Distributed energy resources
- Robotics & Controls

PUBLICATION

Decision Tree Variations and Online Tuning for Real-Time Control of a Building in a Two-Stage Management Strategy, DOI: [10.3390/en17112730](https://doi.org/10.3390/en17112730)

REFERENCES

Rémy Rigo-Mariani (Research Scientist)
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Vanessa Evers (Professor)
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